

Constrained Optimization II

Exercise 1

Two voltage sources connected in series with voltages x_1 [v] and x_2 [v] respectively are required to provide a constant voltage $V_L = 1$ [v] for a load. The cost for using the first one is x_1^2 . The voltage of the first one is upper bounded by 0.5 [v]. The cost of using the second one is $2x_2^2$. The voltage of the second one is not upper bounded.

- (a) Find the candidate solutions using the KKT conditions.
- (b) Are the KKT conditions sufficient for this problem? Justify your answer.